

Daniel Waxman

Basis Research Institute
Cambridge, MA

dan@basis.ai / dwaxman@mit.edu
danwaxman.github.io

EDUCATION

- Ph.D. Electrical Engineering, *December 2025*
Stony Brook University
Thesis: *Sequential Bayesian Learning and Ensembles for Online Inference and Experimental Design*
Advisor: Petar M. Djurić
- B.S. Mathematics & Applied Mathematics and Statistics (*cum laude*), May 2021
Stony Brook University

EMPLOYMENT & ACADEMIC APPOINTMENTS

- 01/26 – Postdoctoral Research Scientist
Present *Basis Research Institute*, Cambridge, MA.
Postdoctoral fellowship jointly supervised by Matthew Levine (Basis) and Youssef Marzouk (MIT).
- 05/26 – Research Affiliate
Present *Laboratory for Information & Decision Systems*
Massachusetts Institute of Technology, Cambridge, MA.
- 06/25 – Research Intern
01/26 *Basis Research Institute*, New York, NY.
Research in causal inference, probabilistic programming, and applied Bayesian machine learning with Rafal Urbaniak and Jack Feser.

RESEARCH INTERESTS

- Bayesian Machine Learning — particularly problems in dynamical systems and online settings
- Ensembling Methods — particularly Bayesian-based ensembling rules and online ensembling
- Causal Inference — particularly causal discovery and causality for dynamical systems

PUBLICATIONS

* denotes equal contribution

Preprints and Articles In Review

- 2026 **D. Waxman**, D. Batenkov, J. Feser, A. Zane, E. Bingham, Y. Marzouk, M. Levine, “Dynestyx: A Probabilistic Programming Library for Dynamical Systems.” *Submitted*.
[\[arXiv\]](#) [\[code\]](#)
- 2026 M. Levine, **D. Waxman**, I. Urteaga. “CD-Dynamax: A JAX-based Python package for continuous-discrete probabilistic state space modeling and inference.” *Submitted*.
[\[pdf\]](#) [\[code\]](#)

- 2026 K. Lamer, E. V. Davis, Z. Mages, Z. Zhu, N. E. Rojas-Robles, K. N. Mitu, E. R. Vivoni, Z. Wang, H. Li, M. Cadeddu, P. Joshi, **D. Waxman**. “The Encanto Neighborhood Field Experiments Dataset: Land–Atmosphere Interactions in Urban Phoenix.” *Submitted*.

Journal Papers

- 2026 **D. Waxman**, F. Llorente, P. M. Djurić, “Sequential Inference with Gaussian Processes: A Signal Processing Perspective.” Accepted to *IEEE Signal Processing Magazine*.
[arXiv]
- 2026 **D. Waxman**, F. Llorente, P. M. Djurić. “Bayesian Ensembling: Insights from Online Optimization and Empirical Bayes.” *Transactions on Machine Learning Research (TMLR)*, 2026.
[OpenReview] [arXiv] [code]
- 2025 **D. Waxman**, F. Llorente, K. Lamer, P. M. Djurić. “Designing an Optimal Sensor Network via Minimizing Information Loss.” *Bayesian Analysis*, 2025.
[ProjectEuclid] [arXiv] [code]
- 2024 **D. Waxman**, P. M. Djurić. “Dynamic Online Ensembles of Basis Expansions.” *Transactions on Machine Learning Research (TMLR)*, 2024.
[OpenReview] [arXiv] [code]
- 2024 **D. Waxman**, K. Butler, P. M. Djurić. “DAGMA-DCE: Interpretable, Non-Parametric Differentiable Causal Discovery.” *IEEE Open Journal of Signal Processing*, vol. 5, pp. 393-401, 2024.
[IEEE Xplore] [arXiv] [code]

Conference Papers

- 2026 **D. Waxman**, M. Levine, K. Butler, P.M. Djurić, E. Mackevicius, D. Batenkov. “Bayesian Learning of Graph-Based Interaction Potentials with Applications to Studying Animal Behavior.” To appear in *Proceedings of the 2026 Asilomar Conference on Signals, Systems, and Computers*, 2026.
- 2026 M. Heinonen, Y. Shenfeld, R. Baptista, **D. Waxman**, D. Batenkov, T. Cooijmans, E. Bingham. “Wasserstein Residuals: Learning Gradient Flows from Population Dynamics.” Accepted to *ICML 2026 Workshop on Structured Probabilistic Inference & Generative Modeling (SPIGM)*.
- 2026 F. Llorente, **D. Waxman**, S. Jantre, N. Urban, S. Minkoff. “Robust, Online, and Adaptive Decentralized Gaussian Processes.” *Proceedings of the 2026 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2026.
[IEEEXplore] [arXiv]
- 2025 **D. Waxman**, P. M. Djurić. “Non-Stationary Causal Learning via Hierarchical Modeling.” *Proceedings of the 2025 Asilomar Conference on Signals, Systems, and Computers*, 2025.
[IEEEXplore]
- 2025 F. Llorente*, **D. Waxman***, P. M. Djurić. “Decentralized Online Ensembles of Gaussian Processes for Multi-Agent Systems.” In *Proceedings of the 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.
[IEEEXplore] [arXiv] [code]
- 2024 K. Butler*, **D. Waxman***, P. M. Djurić. “Tangent Space Causal Inference: Leveraging Vector Fields for Causal Discovery in Dynamical Systems.” In *Advances in Neural Information Processing Systems*, 37, 120078-120102, 2024.
[OpenReview] [arXiv] [code]
- 2024 **D. Waxman**, P. M. Djurić. “A Gaussian Process-based Streaming Algorithm for Prediction of Time Series With Regimes and Outliers.” In *Proceedings of the 27th International Conference on Information Fusion (FUSION)*, Venice, Italy, 2024.
[IEEEXplore] [arXiv] [code]

- 2023 M. Ajirak, **D. Waxman**, F. Llorente, P. M. Djurić. “Fusion of Gaussian Process Predictions with Monte Carlo Sampling.” *Proceedings of the 57th Asilomar Conference on Signals, Systems, and Computers* (pp. 1367-1371), 2023.
[IEEE Xplore] [arXiv]
- 2023 Y. Liu, C. Cui, **D. Waxman**, K. Butler, P. M. Djurić. “Detecting Confounders in Multivariate Time Series Using Strength of Causation.” In *Proceedings of the 31st European Signal Processing Conference (EUSIPCO)*, pp. 1400-1404, 2023.
[IEEE Xplore] [pdf] [video]

Professional Meeting Abstracts

- 2026 K. Lamer, E. V. Davis, Z. J. Mages, P. Joshi, N. Rojas, E. R. Vivoni, Z. Wang, Z. Zhu, **D. Waxman**. “Multisensor Observations and Meter-Scale Modeling of Urban Park Cooling Dynamics for Digital Twin Integration.” *106th AMS Annual Meeting*.
- 2024 D. Chernoff, **D. Waxman**, G. Brite, L. Langman, C Rabeno, L. Czerwinka, P. M. Djurić. “The Development of an Automated Algorithm to Identify and Manage Post-thyroidectomy Hypocalcemia.” *American Head and Neck Society 2024 General Meeting*.

RESEARCH TALKS

- 2026 SIAM New England Section Annual Meeting ’26
Minisymposium on “Generative Scientific Machine Learning for PDEs and Complex Physical Systems”
Talk Title: “A Bayesian Recipe for Learning Hybrid Models from Noisy Data”
- 2026 ISBA World Meeting ’26
Session on “Calibrated Bayes: Model Design and Adaptation Under Limited Resources”
Talk Title: “Online & Predictively-Oriented Model Fusion”
- 2026 SIAM Uncertainty Quantification ’26
Minisymposium on “Sequential Inference for Dynamical Systems”
Talk Title: “GP4SP: Sequential Gaussian Process Inference for Signal Processing”
- 2026 Queens College Computer Science Colloquium
Talk Title: “Online Bayesian Learning and Ensembles” [slides]
- 2024 SIAM New York-New Jersey-Pennsylvania Section Conference ’24
Talk Title: “Optimizing Observation Locations via Minimizing Information Loss”
- 2023 Acoustics Research Institute of the Austrian Academy of Sciences
Talk Title: “Causal Discovery via Quantifying Influences” [link] [slides]
- 2023 Bellairs Workshop on Machine Learning and Statistical Signal Processing for Data on Graphs
Talk Title: “Bayesian Combination”

SELECTED HONORS AND AWARDS

- 2026 Best Poster Award @ Bayesian Young Statisticians Meeting (BAYSM) 2026
- 2026 Gold Reviewer @ ICML 2026
- 2025 Top Reviewer @ NeurIPS 2025

TEACHING

As Instructor of Record

Fall 2025 Understanding Machine Learning (ESE 188 @ SBU)

Spr 2025 Understanding Machine Learning (ESE 188 @ SBU)

As Teaching Assistant

Sum 2022 Random Signals and Systems (ESE 306 @ SBU)

Spr 2022 Random Signals and Systems (ESE 306 @ SBU)

Fall 2021 Programming Fundamentals (ESE 124 @ SBU)

OUTREACH & SERVICE

Reviewing

I've served as a reviewer for 60+ manuscripts across top-tier journals and conferences in machine learning and signal processing, including several top reviewer awards.

Journals IEEE Transactions on Signal Processing [2026]
IEEE Transactions on Aerospace and Electronic Systems [2026]
Transactions on Machine Learning Research (TMLR) [2024, 2025, 2026]
Journal of Open Source Software (JOSS) [2024, 2025, 2026]
EURASIP Journal on Advances in Signal Processing [2023]

Conf. NeurIPS [2025; **Top Reviewer** 2025]
ICLR [2025, 2026]
ICML [2025, **Position Track Gold Reviewer** 2026]
European Signal Processing Conference (EUSIPCO) [2024, 2025, 2026]
NeurIPS Workshops [2024]
IEEE Statistical Signal Processing Workshop [2025]
Asilomar Conferences on Signals, Systems, and Computers (ASSC) [2025, 2026]
AAAI [2026]

University Service

2021–25 Stony Brook University Graduate Student Organization
Primary Senator, Dept. of Electrical & Computer Engineering
Graduate Student Representative for Graduate Council
Reviewer for Distinguished Travel Award

2022–23 SBU Strategic Planning Committee [\[link\]](#)
Member of Strategic Planning Committee; helped draft vision statement for Project REACH

Expository Talks

2024 SBU ECE Honors Seminar
Ensembling and Fusion: Perspectives and Solutions from Signal Processing
2022 Stony Brook Math Club
Making Markov Chains with Metropolis [\[video\]](#)

Organizing

- 2026 Special Session on “Bayesian Learning and Inference on Graphs”
Co-organized with Kurt Butler, Victor Elvira, and Petar M. Djurić
Asilomar 2026, Pacific Grove, CA
- 2026 Minisymposium on “Sequential Inference for Dynamical Systems”
Co-organized with Fernando Llorente-Fernandez
SIAM UQ 2026, Minneapolis, MN.
- 2025 Special Session on “Advances in Causal Inference: Theory and Applications”
Co-organized with Petar M. Djurić
Asilomar 2025, Pacific Grove, CA

Mentoring

- 2022–25 CUNY Directed Reading Program
Mike Prebil (*Gaussian Processes and Bayesian Optimization*)
Edgar Cuapio Diaz (*Gaussian Processes and Bayesian Optimization*)
Jonathan Jaimangal (*Hamiltonian Monte Carlo*, **Winner Outstanding Poster Award**)
Masroor Khonkhodzhaev (*Dynamic Programming and Reinforcement Learning*)
Percy Martinez (*Fourier Analysis*)
Isabella Chittumuri (*Elements of Statistical Learning*)
- 2023 Stony Brook DRP
Shailen Smith (*Probabilistic Machine Learning*)
- 2022 Bayesian Data Analysis for the Global South
Volunteer TA for online course by Aki Vehtari [[certificate](#)]

Updated June 2026